Sri Lanka Institute of Information Technology



Group Report

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Computer Networks – IE2004

B.Sc. (Hons) in Computer Science / B.Sc. (Hons) in Computer Systems Engineering

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Introduction

This report shows a full network infrastructure design for a small IT business that works out of a two-story commercial building in Sri Lanka. The design includes analyzing requirements, creating a logical network architecture, and planning the physical infrastructure. It follows industry standards and international cabling standards.

❖ **ENTERPRISE OVERVIEW**

- **Company Profile**

  - **Company Name:** Software Solutions (Pvt) Ltd
  - **Industry:** Information Technology Services
  - **Business Type:** Software Development & IT Consulting
  - **Registration:** Private Limited Company, Sri Lanka
  - **Establishment Year:** 2023
  - **Company Registration No:** PV 1928374
  - **Business Address:** Software Solutions (Pvt) Ltd. No. 456, Galle Road, Colombo 03, Sri Lanka

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❖ **Business Activities**

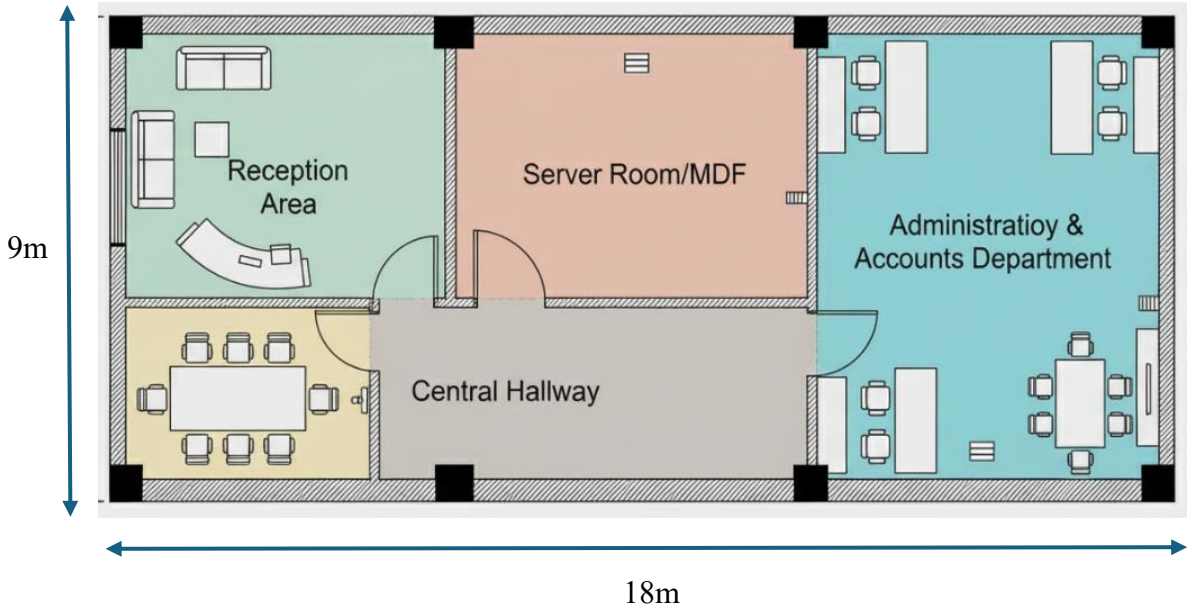
Core Services:

1. **Custom Software Development**
   - Enterprise applications
   - Mobile application development (iOS/Android)
   - Web application development
   - Database design and implementation
2. **IT Consulting Services**
   - IT infrastructure planning
   - Digital transformation consulting
   - Business process automation
   - Technology advisory
3. **Cloud Solutions**
   - Cloud migration services
   - AWS/Azure infrastructure setup
   - Cloud application hosting
   - DevOps implementation
4. **Managed IT Services**
   - Network administration
   - Server maintenance
   - Helpdesk support
   - Cybersecurity services

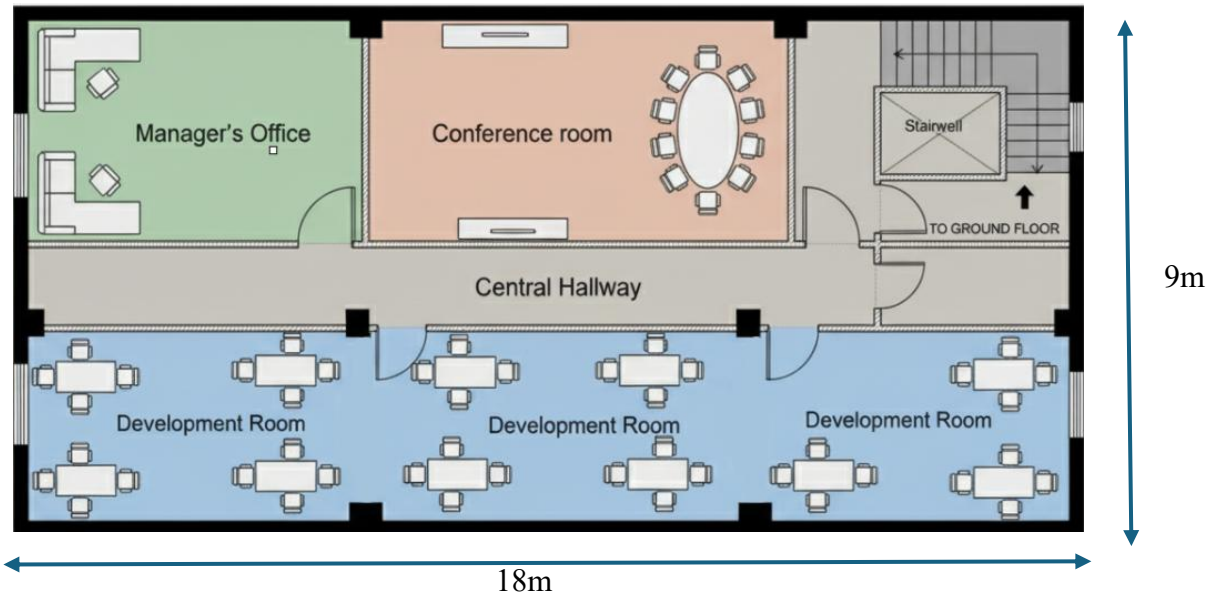
1 Requirement Analysis

1.a Floor Plans with Dimensions

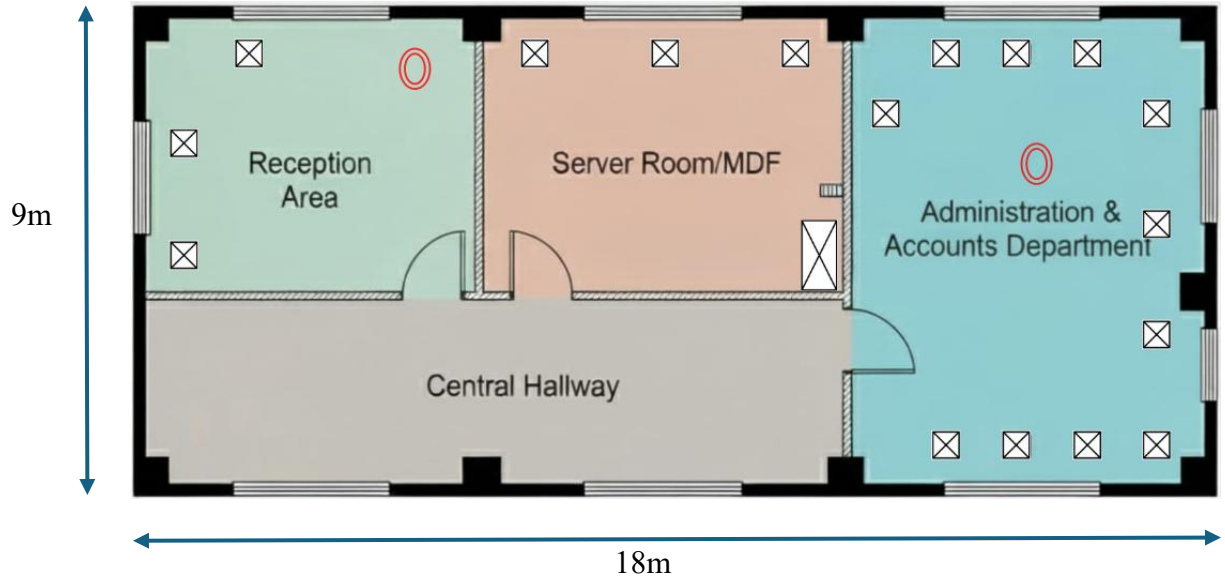
Ground Floor Plan
Building Size: 18.0m × 9.0m = 162 m²
Scale: 1:100



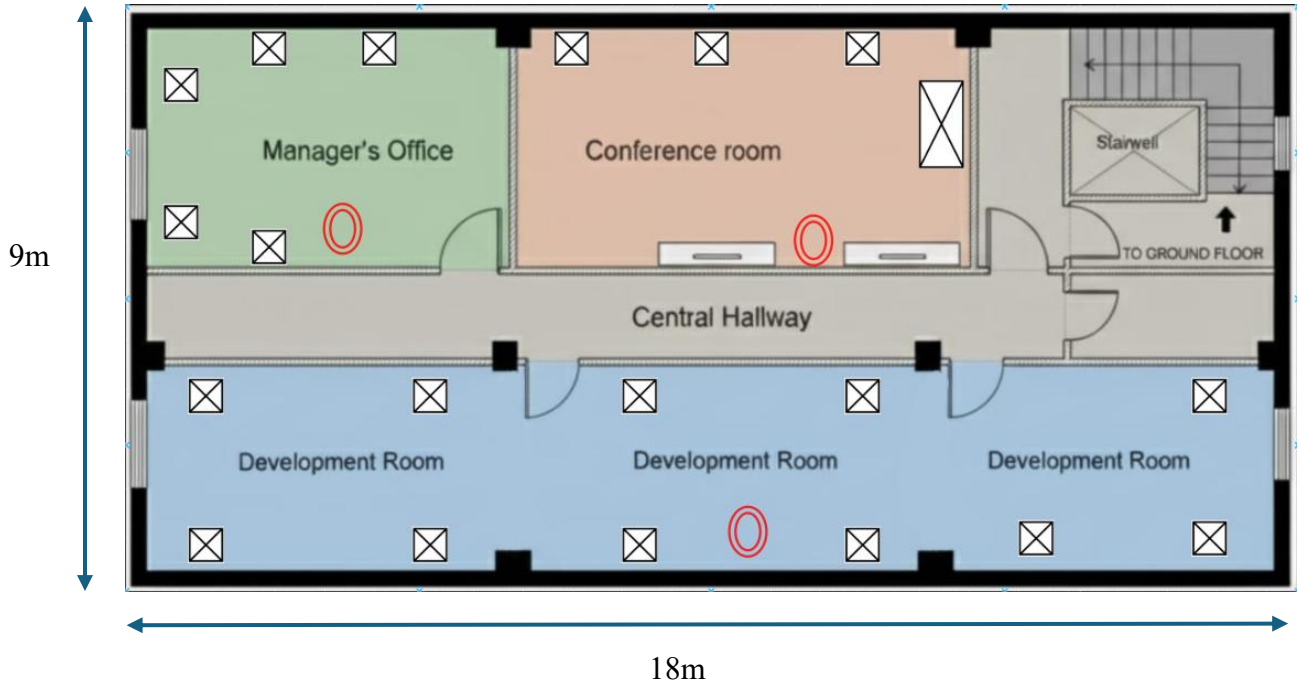
First Floor Plan
Building Size: 18.0m × 9.0m = 162 m²
Scale: 1:100



Ground Floor Plan with Locations of end devices (hosts)



First Floor Plan with Locations of end devices (hosts)



DATA OUTLET

EQUIPMENT RACK

ACCESS POINT

Table - Complete Wired Device Inventory

Floor	Room	Desktop PCs	Printers	IP Phones	Wireless Access Points	Server	Total
Ground Floor	Reception	1	1	1	1	0	4
	Server Room	1	0	1	0	1	3
	Administration	8	2	1	1	0	12
First Floor	Manager's Office	3	1	1	1	0	5
	Conference Room	1	1	1	1	0	4
	Development	8	2	1	1	0	12
Total		**22**	**7**	**6**	**5**	**1**	**41**

Wired End Device Summary:

- Desktop PCs: 22
- Network Printers: 7
- IP Phones (VoIP): 6
- Wireless Access Points: 5
- Servers: 1
- **Total Wired Devices: 41**

1.b End Device Locations - Wired Connectivity

Table - Complete Wired Device Inventory by Location

Floor	Room	Device Type	Quantity	Device IDs	Connection Type	Purpose
Ground Floor	**Server Room**	**Router**	**1**	**ISR-Router**	**Rack-mounted**	**Internet gateway**
		Layer 3 Switches	**2**	**Core-Switch-01, Switch-GF**	**Rack-mounted**	**Network backbone**
		Layer 2 Switch	**1**	**SW-Server Room**	**Rack-mounted**	**Room connectivity**
		Server	**1**	**Server-01**	**Rack-mounted**	**Application server**
		Desktop PC	**1**	**PC-Server-01**	**Wall outlet**	**Management**

Floor	Room	Device Type	Quantity	Device IDs	Connection Type	Purpose
		IP Phone	1	Phone-Server	Wall outlet	Communication
	Reception	Desktop PC	1	PC-Reception	Wall outlet	Front desk
		Network Printer	1	Printer-Reception	Wall outlet	Reception printing
		IP Phone	1	Phone-Reception	Wall outlet	Reception calls
		Wireless AP	1	AP-Reception	Ceiling outlet (PoE)	Wi-Fi coverage
	Admin	Desktop PC	8	PC-Admin-01 to -08	Wall outlet	Office workstations
		Network Printer	2	Printer-Admin-01, -02	Wall outlet	Department printing
		IP Phone	1	Phone-Admin	Wall outlet	Dept communication
		Wireless AP	1	AP-Admin	Ceiling outlet (PoE)	Wi-Fi coverage
First Floor	Manager's Office	Desktop PC	3	PC-Manager-01, -02, -03	Wall outlet	Management work
		Network Printer	1	Printer-Managers	Wall outlet	Office printing
		Wireless AP	1	AP-Manager	Ceiling outlet (PoE)	Wi-Fi coverage
		IP Phone	1	Phone-Managers	Wall outlet	Management calls
	Conference Room	Desktop PC	1	PC-Conference	Floor box	Presentations

Floor	Room	Device Type	Quantity	Device IDs	Connection Type	Purpose
		Network Printer	1	Printer-Conference	Wall outlet	Meeting materials
		IP Phone	1	Phone-Conference	Floor box	Conference calls
		Wireless AP	1	AP-Conference	Ceiling outlet (PoE)	Wi-Fi coverage
	Developer Room	Desktop PC	8	PC-Dev-01 to -08	Wall outlets	Software development
		Network Printer	2	Printer-Dev-01, -02	Wall outlets	Code printing
		IP Phone	1	Phone-Dev	Wall outlet	Team communication
		Wireless AP	1	AP-Dev	Ceiling outlet (PoE)	Wi-Fi coverage

PoE – Power over Ethernet

Total Wired End Devices: 41
Network Infrastructure:

- 1 Router (ISR4331)
- 9 Switches
- 5 Wireless Access Points

1.c End Device Locations - Wireless Connectivity

Wireless Access Point Deployment

Table - Access Point Coverage Analysis

AP Name	Location	Floor	Room	Channel (2.4G/5G)	Expected Users	Backhaul
AP-Reception	Ceiling center	Ground	Reception	1 / 36	5-10	100 Mbps PoE
AP-Admin	Ceiling center	Ground	Administration	6 / 48	12-18	100 Mbps PoE

AP Name	Location	Floor	Room	Channel (2.4G/5G)	Expected Users	Backhaul
AP-Conference	Ceiling center	First	Conference Room	11 / 149	8-15	100 Mbps PoE
AP-Dev	Ceiling center	First	Development	1 / 64	15-20	100 Mbps PoE
AP-Manager	Ceiling center	First	Managers	11 / 161	3-8	100 Mbps PoE

1.d Internet Connectivity

Internet Service Provider: Sri Lanka Telecom (SLT)
Service Package: SLT Fiber Business 100
Technology: GPON (Gigabit Passive Optical Network) Fiber to the Premises (FTTH)
Connection Specifications:

- **Download Speed:** 100 Mbps (guaranteed minimum)
- **Upload Speed:** 100 Mbps (symmetric connection)
- **Connection Type:** Ethernet handoff from ISP-provided ONT (Optical Network Terminal)
- **Data Allowance:** Unlimited (Fair Usage Policy applies)
- **Public IP Address:** 1 Static IPv4 address
- **IPv6 Support:** /64 subnet allocation
- **Support:** 24/7 hotline (1212), 4-hour response time

Physical Connection: The fiber service terminates at an ONT (Huawei HG8546M or equivalent) mounted in the Server Room at 1.8m above floor level on the south wall. The ONT provides an Ethernet handoff via RJ45 connector, which connects to the enterprise router's WAN interface (ISR4331 Gi0/0/1) using a 1.5-meter CAT6 patch cable.

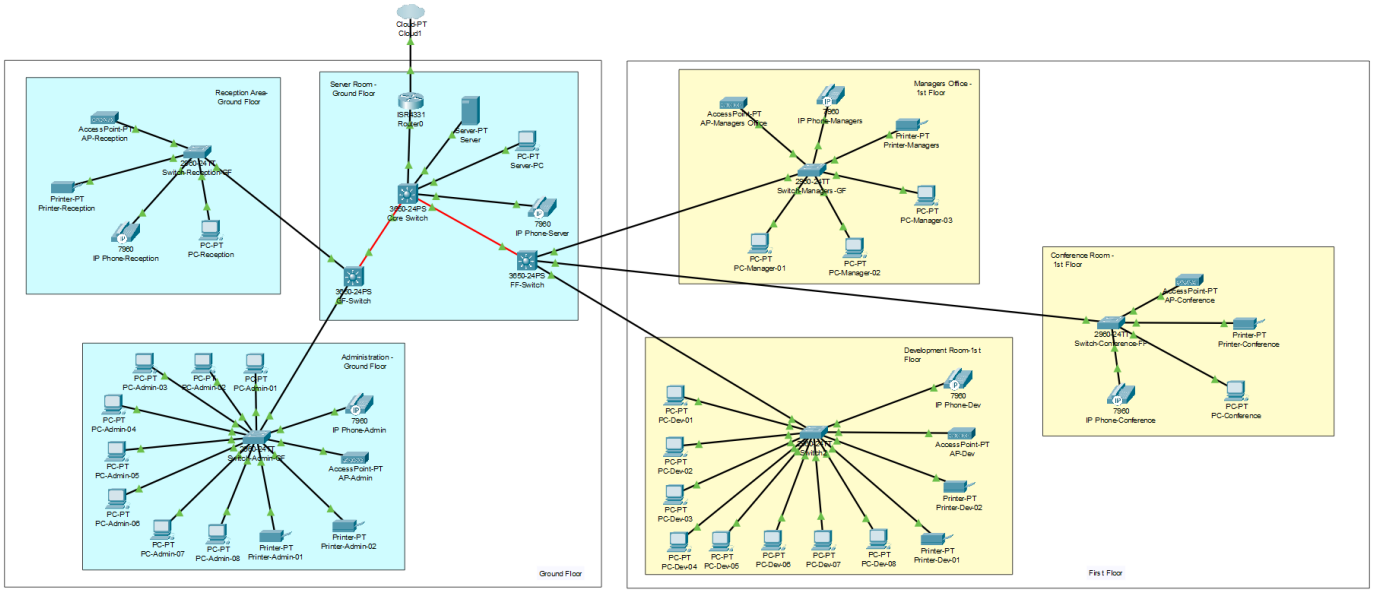
Cost Analysis:

- **Monthly Subscription:** LKR 12,000
- **Static IP Add-on:** LKR 1,500/month
- **Installation Fee:** LKR 15,000 one-time
- **First Year Total:** LKR 177,000
- **Subsequent Years:** LKR 162,000 annually

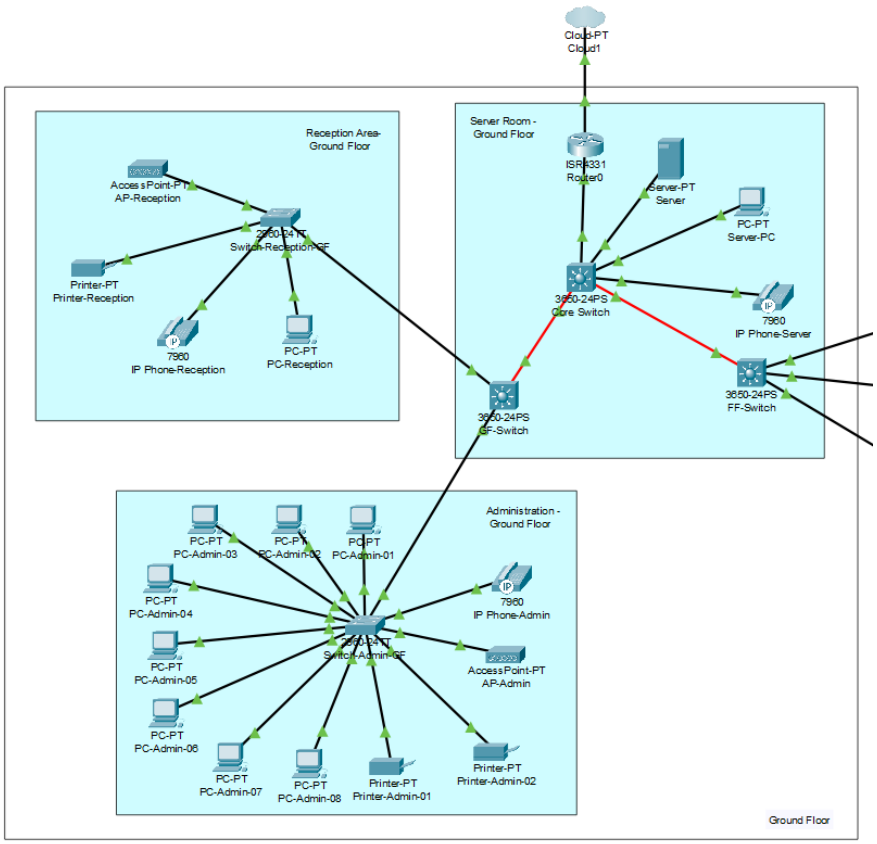
The 100 Mbps symmetric fiber connection provides adequate bandwidth for the enterprise's 40 employees, supporting concurrent VoIP calls, video conferencing, cloud services, and general internet usage with 25% headroom for peak periods.

2 Logical Network Design

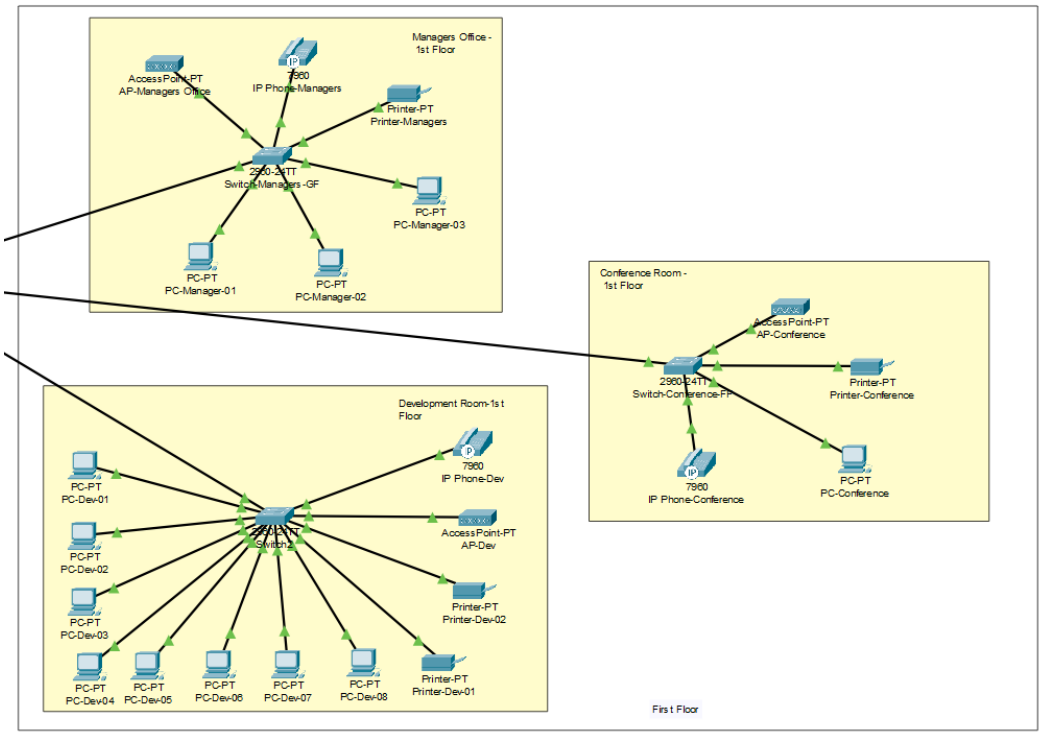
2.a Network Topology Diagram



Ground Floor



First Floor



2.b Equipment List

Device	Model	Type	Location	Ports Used
ISR-Router	Cisco ISR4331	Router	Server Room (GF)	2
Core-Switch-01	Catalyst 3650-24PS	L3 Switch	Server Room (GF)	5
Switch-GF	Catalyst 3650-24PS	L3 Switch	Server Room (GF)	3
Switch-FF	Catalyst 3650-24PS	L3 Switch	Conference Room (FF)	4
SW-Reception	Catalyst 2960-24TT	L2 Switch	Server Room (GF)	5
SW-Admin	Catalyst 2960-24TT	L2 Switch	Server Room (GF)	13
SW-Managers	Catalyst 2960-24TT	L2 Switch	Conference Room (FF)	7
SW-Conference	Catalyst 2960-24TT	L2 Switch	Conference Room (FF)	5
SW-Dev	Catalyst 2960-24TT	L2 Switch	Conference Room (FF)	13
AP-Reception	Wireless AP	Access Point	Reception (GF)	1
AP-Admin	Wireless AP	Access Point	Administration (GF)	1
AP-Conference	Wireless AP	Access Point	Conference (FF)	1
AP-Dev	Wireless AP	Access Point	Development (FF)	1
AP-Managers	Wireless AP	Access Point	Managers (FF)	1

The Catalyst 3650-24PS contains GLC-LH-SMD module.

Ground Floor Switches:

- SW-Reception
- SW-Admin

First Floor Switches:

- SW-Managers
- SW-Conference
- SW-Dev

2.c Link Speeds

Table - Infrastructure Link Speeds

Connection	Cable Type	Speed	Duplex
Internet to Router	CAT6 UTP	100 Mbps	Full
Router to Core Switch	CAT6 UTP	1 Gbps	Full
Core to Switch-GF	OM3 Fiber	1 Gbps	Full
Core to Switch-FF	OM3 Fiber	1 Gbps	Full
Switch-GF to Room Switches	CAT6 UTP	1 Gbps	Full
Switch-FF to Room Switches	CAT6 UTP	1 Gbps	Full
Room Switches to End Devices	CAT6 UTP	100 Mbps	Full (auto)

Key Points:

- All infrastructure links: 1 Gbps (Gigabit Ethernet)
- Fiber for inter-floor: OM3 Multimode, 1 Gbps
- End devices: 100 Mbps Fast Ethernet with auto-negotiation.
- All links: Full-duplex operation

All the network switches follow the IEEE 802.1Q standard, which lets us set up VLANs for future network segmentation and traffic management. The Catalyst 3650-24PS switches (Core and Distribution layers) and Catalyst 2960-24TT switches (Access layer) which let us logically separate the network by department, function, or security need without having to make any changes to the physical infrastructure.

3 Physical Network Design - Passive Components

3.a UTP Patch Panels

Table : UTP Patch Panel Specifications and Port Assignments

Panel ID	Rack Location	Port Count	Cable Category	Ports Used
PP-GF-01	RACK-GF	**24-port**	CAT6 UTP	5
PP-GF-02	RACK-GF	**24-port**	CAT6 UTP	20
PP-FF-01	RACK-FF	**24-port**	CAT6 UTP	13
PP-FF-02	RACK-FF	**24-port**	CAT6 UTP	20

UTP Patch Panel Summary:

- Total Panels: 4 (2 per floor)
- Total Port Capacity: 96 ports (4 panels × 24 ports)
- Ports Currently Used: 58 ports
- Port Count per Panel: 24 ports

All UTP Patch Panels:

- **Format:** 1U height, 19-inch rackmount
- **Port Configuration:** 24× RJ45 female jacks (front), 110-punch down blocks (rear)
- **Cable Category:** CAT6 UTP (TIA-568-C.2 compliant)
- **Bandwidth:** 250 MHz
- **Speed Support:** 10/100/1000BASE-T (up to 1 Gbps)
- **Wiring Standard:** T568B pin configuration
- **Port Numbering:** Left to right, 1-24.

3.b Fiber Patch Panels: Port Count and Connector Type

Table - Fiber Patch Panel Specifications

Panel ID	Rack Location	Floor	Rack U Position	Port Count	Connector Type	Fiber Type	Ports Used
FP-GF-01	RACK-GF	Ground Floor	U34	**12 ports**	**LC duplex**	OM3 MM 50/125µm	2 (1 duplex)
FP-FF-01	RACK-FF	First Floor	U9	**12 ports**	**LC duplex**	OM3 MM 50/125µm	2 (1 duplex)

Fiber Patch Panels:

- **Total Panels:** 2 (1 per floor)
- **Total Port Count:** 24 ports (12 ports per panel)
- **Connector Type:** LC duplex (push-pull connectors)
- **Duplex Connections:** 12 duplex connections total (6 per panel)
- **Ports Currently Used:** 4 ports (2 duplex connections)

Port Count Details:

- **Physical Ports:** 12 LC connectors per panel
- **Duplex Pairs:** 6 duplex connections (each duplex = 2 physical ports)
- **Active Duplex:** 1 per panel (connecting ground floor to first floor)
- **Reserved Duplex:** 5 per panel (for redundancy and future expansion)

Connector Type - LC Duplex:

- **Form Factor:** LC (Lucent Connector) - small form factor
- **Configuration:** Duplex (2 fibers per connector: transmit + receive)
- **Polarity:** Method A (straight-through)
- **Color:** Blue housing (standard for single-mode compatible, works with multimode)
- **Industry Standard:** Most common connector for enterprise networks

Fiber Type:

- **Classification:** OM3 Multimode
- **Core/Cladding:** 50/125 micrometers
- **Jacket Color:** Aqua (OM3 standard color code)
- **Bandwidth:** 2000 MHz - km @ 850nm
- **Speed Support:** 1 Gbps (up to 1000m), 10 Gbps (up to 300m)

3.c Wall Outlets: Locations

Ground Floor - Wall Outlets and Infrastructure

Table - Ground Floor Wall Outlets

Outlet ID	Location	Wall/Position	Height	Port Configuration	Cable Length
WO-GF-01	Reception	West wall	0.3m AFF	2× RJ45 (dual port)	6m
WO-GF-02	Reception	North wall	0.3m AFF	1× RJ45 (single port)	7m
WO-GF-03	Reception	West wall	0.3m AFF	1× RJ45 (single port)	6m
WO-GF-04	Reception	Ceiling center	2.8m AFF	1× RJ45 (PoE)	8m

Outlet ID	Location	Wall/Position	Height	Port Configuration	Cable Length
WO-GF-05	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	17m
WO-GF-06	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	20m
WO-GF-07	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	24m
WO-GF-08	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	27m
WO-GF-09	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	22m
WO-GF-10	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	25m
WO-GF-11	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	29m
WO-GF-12	Administration	South wall	0.3m AFF	2× RJ45 (dual port)	32m
WO-GF-13	Administration	South wall	0.3m AFF	1× RJ45 (single port)	20m
WO-GF-14	Administration	South wall	0.3m AFF	1× RJ45 (single port)	30m
WO-GF-15	Administration	South wall	0.3m AFF	1× RJ45 (single port)	26m
WO-GF-16	Administration	Ceiling center	2.8m AFF	1× RJ45 (PoE)	28m

Ground Floor Wall Outlet Summary:

- **Total Outlets:** 16 locations
- **Total Ports:** 24 RJ45 ports (12 dual-port + 4 single-port outlets)
- **Dual-Port Outlets:** 10 (for PCs with phones)
- **Single-Port Outlets:** 4 (for printers/dedicated devices)
- **Ceiling Outlets:** 2 (for wireless APs, PoE capable)
- **Total Cable Length:** ~330 meters CAT6 UTP

First Floor - Wall Outlets and Infrastructure

Table - First Floor Wall Outlets

Outlet ID	Location	Wall/Position	Height	Port Configuration	Cable Length
WO-FF-17	Manager's Office	West wall	0.3m AFF	2× RJ45 (dual port)	5m
WO-FF-18	Manager's Office	West wall	0.3m AFF	2× RJ45 (dual port)	6m
WO-FF-19	Manager's Office	South wall	0.3m AFF	2× RJ45 (dual port)	7m
WO-FF-20	Manager's Office	West wall	0.3m AFF	1× RJ45 (single port)	5m
WO-FF-21	Managers	Ceiling center	2.8m AFF	1× RJ45 (PoE)	13m
WO-FF-22	Conference Room	West wall	0.3m AFF	2× RJ45 (dual port)	4m
WO-FF-23	Conference Room	Floor box	Floor level	2× RJ45 (dual port)	6m
WO-FF-24	Conference Room	North wall	0.3m AFF	1× RJ45 (single port)	5m
WO-FF-25	Conference Room	Ceiling center	2.8m AFF	1× RJ45 (PoE)	6m
WO-FF-26	Development	South wall	0.3m AFF	2× RJ45 (dual port)	14m
WO-FF-27	Development	South wall	0.3m AFF	2× RJ45 (dual port)	18m
WO-FF-28	Development	South wall	0.3m AFF	2× RJ45 (dual port)	22m
WO-FF-29	Development	South wall	0.3m AFF	2× RJ45 (dual port)	26m
WO-FF-30	Development	South wall	0.3m AFF	2× RJ45 (dual port)	19m
WO-FF-31	Development	South wall	0.3m AFF	2× RJ45 (dual port)	23m

Outlet ID	Location	Wall/Position	Height	Port Configuration	Cable Length
WO-FF-32	Development	South wall	0.3m AFF	2× RJ45 (dual port)	27m
WO-FF-33	Development	South wall	0.3m AFF	2× RJ45 (dual port)	31m
WO-FF-34	Development	South wall	0.3m AFF	1× RJ45 (single port)	24m
WO-FF-35	Development	South wall	0.3m AFF	1× RJ45 (single port)	30m
WO-FF-36	Development	South wall	0.3m AFF	1× RJ45 (single port)	22m
WO-FF-37	Development	Ceiling center	2.8m AFF	1× RJ45 (PoE)	27m

AFF – Above First Floor

First Floor Wall Outlet Summary:

- **Total Outlets:** 20 locations
- **Total Ports:** 29 RJ45 ports (13 dual ports + 4 single port + 1 floor box)
- **Dual-Port Outlets:** 13 (workstations)
- **Single-Port Outlets:** 4 (printers/dedicated)
- **Floor Box:** 1 (conference table, dual port)
- **Ceiling Outlets:** 3 (wireless APs, PoE)
- **Total Cable Length:** ~350 meters CAT6 UTP

3.d Equipment Racks: Location, Type, Height

Table - Equipment Rack Specifications

Rack ID	Location	Floor	Type	Height	Dimensions (W×D)
RACK-GF	Server Room	Ground Floor	**Floor-standing**	**42U** (1867mm / 73.5")	600mm × 1000mm
RACK-FF	Conference Room	First Floor	**Wall-mounted**	**12U** (533mm / 21")	600mm × 600mm

Equipment Rack Summary:

- **Total Racks:** 2 (1 per floor)
- **Ground Floor:** 42U floor-standing (MDF - Main Distribution Frame)
- **First Floor:** 12U wall-mounted (IDF - Intermediate Distribution Frame)

Detailed Rack Specifications

- **RACK-GF (Ground Floor - Main Distribution Frame)**

Location Details:

- **Building:** Ground Floor
- **Room:** Server Room (5.5m × 4m, 22 m²)
- **Exact Position:** 0.5 meters from north wall, positioned against west side of room
- **Distance from door:** 2 meters (allows clear access)
- **Clearance:** 1-meter front clearance, 0.8-meter rear clearance for maintenance

Type: Floor-standing 19-inch 4-post equipment rack

Height: 42U

- **U Measurement:** 1U = 44.45mm (1.75 inches)
- **Total Height:** 42U × 44.45mm = 1867mm (73.5 inches)
- **With Frame:** Approximately 2000mm (2 meters) total height
- **EIA-310-D Compliant:** Standard 19-inch rack unit spacing.

Physical Specifications:

- **External Width:** 600mm - standard 19-inch rack
- **External Depth:** 1000mm.
- **Mounting Rails:** 4-post adjustable square-hole rails (M6 cage nuts).
- **Usable Depth:** 800mm between front and rear rails.

- **RACK-FF (First Floor - IDF)**

Location Details

- **Building: First Floor**
- **Room: Conference Room /** IDF **(6.5m × 4m, 26 m²)**
- **Exact Position: Mounted on east wall, 1.5 meters from south corner**
- **Height Above Floor: 1.5 meters to bottom of rack (optimal for standing access)**
- **Distance from Door: 3 meters (clear access for maintenance)**

Height: 12U

- **U Measurement:** 1U = 44.45mm (1.75 inches)
- **Total Height:** 12U × 44.45mm = 533mm (21 inches)
- **With Frame:** Approximately 600mm total height (including mounting brackets)
- **EIA-310-D Compliant:** Standard 19-inch rack unit spacing

Physical Specifications:

- **External Width:** 400mm – standard rack
- **External Depth:** 700mm.
- **Mounting Rails:** 4-post adjustable square-hole rails (M6 cage nuts)
- **Usable Depth:** 600mm between front and rear rails

4 Physical Network Design - Active Components

4.a Switches: specifications of selected products

Table - Switches Deployment and Specifications

Location	Role	Model	Port Count	PoE	Uplink	Purpose
Server Room, GF	Core	Cisco Catalyst 3650-24PS	24× 1G + 4× SFP	370W PoE+	1G fiber/copper	Network core, inter-VLAN routing capability, connect router and distribution switches for centralized traffic management
Server Room, GF	Distribution	Cisco Catalyst 3650-24PS	24× 1G + 4× SFP	370W PoE+	1G fiber to core	Ground floor distribution switch, aggregates SW-Reception, SW-Admin, SW-Server Room access switches
Conference Room, FF	Distribution	Cisco Catalyst 3650-24PS	24× 1G + 4× SFP	370W PoE+	1G fiber to core	First floor distribution switch, aggregates SW-Managers, SW-Conference, SW-Dev access switches
Server Room, GF	Access	Cisco Catalyst 2960-24TT	24× 1G + 2× SFP	No PoE	1G copper to Switch-GF	Reception area connectivity (PC, printer, phone, AP uplink)
Server Room, GF	Access	Cisco Catalyst 2960-24TT	24× 1G + 2× SFP	No PoE	1G copper to Switch-GF	Administration department (8 PCs, 2 printers, phone, AP uplink)
Server Room, GF	Access	Cisco Catalyst 2960-24TT	24× 1G + 2× SFP	No PoE	1G copper to Switch-GF	Server room devices (server, management PC, phone)
Conference Room, FF	Access	Cisco Catalyst 2960-24TT	24× 1G + 2× SFP	No PoE	1G copper to Switch-FF	Manager's office (3 PCs, printer, phone, AP uplink)

Location	Role	Model	Port Count	PoE	Uplink	Purpose
Conference Room, FF	Access	Cisco Catalyst 2960-24TT	24× 1G + 2× SFP	No PoE	1G copper to Switch-FF	Conference room devices (PC, printer, phone, floor box, AP uplink)
Conference Room, FF	Access	Cisco Catalyst 2960-24TT	24× 1G + 2× SFP	No PoE	1G copper to Switch-FF	Development area (8 PCs, 2 printers, phone, AP uplink)

Network Summary:

- **Total Switches:** 9 (3× L3 Distribution/Core + 6× L2 Access)
- **Total Port Capacity:** 244 ports
- **Ports Used:** 60 ports
- **IEEE 802.1Q Compliance:** All switches support VLAN tagging

Cisco Catalyst 3650-24PS

Technical Specifications

Model: WS-C3650-24PS-S

Feature	Specification
Form Factor	1U rackmount, 19-inch EIA standard
Switching Layer	Layer 3 (static routing, OSPF, EIGRP)
Ports	24× 10/100/1000BASE-T + 4× 1G SFP uplink
Switching Capacity	88 Gbps full duplex
Forwarding Rate	65.5 Mpps (million packets per second)
VLAN Support (IEEE 802.1Q)	Yes, up to 4,096 VLANs
PoE+ (IEEE 802.3at)	Yes, 370W total budget, 30W per port
Management	CLI (SSH/console), Web UI, SNMP v3, Syslog
Stacking	StackWise-160 (up to 9 switches, 160 Gbps)
Power Consumption	800W max (370W PoE + 430W system)
Dimensions (H×W×D)	44mm × 445mm × 450mm
MTBF	200,000 hours

Justification for Selection:

- Layer 3 capability enables future inter-VLAN routing without additional equipment
- PoE+ powers all 5 wireless access points (5 APs × 25W = 125W, well under 370W budget)
- 4× SFP uplinks support fiber inter-floor connections (OM3 fiber)

- IEEE 802.1Q compliance meets assignment VLAN requirement
- High switching capacity (88 Gbps) ensures no backbone bottlenecks

Cisco Catalyst 2960-24TT

Technical Specifications

Model: WS-C2960-24TT-L

Feature	Specification
Form Factor	1U rackmount, 19-inch EIA standard
Switching Layer	Layer 2 (VLAN trunking only, no routing)
Ports	24× 10/100/1000BASE-T + 2× 1G SFP uplink
Switching Capacity	52 Gbps full duplex
Forwarding Rate	38.7 Mpps
VLAN Support (IEEE 802.1Q)	Yes, up to 256 VLANs
PoE	No (non-PoE model, reduces cost)
Management	CLI (SSH/console), Web UI, SNMP v3
Power Consumption	60W typical operation
Dimensions (H×W×D)	44mm × 445mm × 368mm
Weight	4.5 kg
MTBF	180,000 hours
Warranty	Cisco Enhanced Limited Lifetime Warranty

Justification for Selection:

- Cost-effective access layer solution (PoE not needed, APs powered via distribution switches)
- 52 Gbps switching capacity adequate for access layer traffic
- IEEE 802.1Q compliance meets assignment requirement
- Low power consumption (60W) reduces UPS sizing and operating costs
- Proven reliability for enterprise access layer deployments

4.b Access Points: Specifications of selected products

Table - Wireless Access Points Deployment and Specifications

Location	Model	Wi-Fi Standard	PoE	Est. Concurrent Clients	Purpose
Reception, Ceiling (GF)	Cisco Aironet 1832i	IEEE 802.11ac Wave 2	IEEE 802.3af (15W)	5-10	Guest and reception wireless coverage, dual-band 2.4 GHz/5 GHz, channel 1/36
Administration, Ceiling (GF)	Cisco Aironet 1832i	IEEE 802.11ac Wave 2	IEEE 802.3af (15W)	12-18	Staff wireless access for administration employees and office guests, channel 6/48
Manager's Office, Ceiling (FF)	Cisco Aironet 1832i	IEEE 802.11ac Wave 2	IEEE 802.3af (15W)	3-8	Management private office wireless coverage, channel 11/161
Conference Room, Ceiling (FF)	Cisco Aironet 1832i	IEEE 802.11ac Wave 2	IEEE 802.3af (15W)	8-15	Meeting room wireless for presentations, video conferencing, and client demos, channel 11/149
Development Area, Ceiling (FF)	Cisco Aironet 1832i	IEEE 802.11ac Wave 2	IEEE 802.3af (15W)	15-20	Developer wireless access, high-density user area, channel 1/64

Wireless Network Summary:

- **Total Access Points:** 5
- **Total Capacity:** 60+
- **Coverage Area:** ~220 m²
- **Average Load:** 8-13 clients per AP

Cisco Aironet 1832i

Technical Specifications

Model: AIR-AP1832I-E-K9

Feature	Specification
Wi-Fi Standard	IEEE 802.11ac Wave 2 (Wi-Fi 5)
Dual Band	Simultaneous 2.4 GHz + 5 GHz operation
Max Data Rate (2.4 GHz)	400 Mbps (2×2 MIMO, 40 MHz channel)
Max Data Rate (5 GHz)	867 Mbps (2×2 MIMO, 80 MHz channel)

Feature	Specification
Total PHY Rate	1.27 Gbps aggregate theoretical throughput
Spatial Streams	2×2:2 (transmit × receive spatial streams)
MU-MIMO	Yes (Multi-User MIMO, simultaneous transmission to 2 clients)
Maximum Clients	100+ supported (50-75 recommended per AP for optimal performance)
Channel Width	20/40 MHz (2.4 GHz), 20/40/80 MHz (5 GHz)
Transmit Power	23 dBm maximum (200 mW), adjustable
Receiver Sensitivity	-96 dBm @ 6 Mbps (best case)
Antennas	4× internal omnidirectional antennas
Coverage Radius	15-20 meters indoors (depends on wall construction)
Security	WPA2-PSK, WPA3-SAE, IEEE 802.1X, AES-CCMP encryption
SSID Support	Up to 16 SSIDs (8 per radio)
Band Steering	Yes (directs dual-band clients to less-congested 5 GHz)
Fast Roaming	IEEE 802.11r/k/v support for seamless AP handoff
Ethernet Port	1× 10/100/1000BASE-T (RJ45)
PoE	IEEE 802.3af (15.4W max), 12.5W typical consumption
Mounting	Ceiling T-bar clips (included), wall mount optional
Dimensions	160mm diameter × 40mm height (low-profile design)
Operating Temperature	0°C to 40°C
MTBF	150,000 hours

Justification for Selection:

- IEEE 802.11ac Wave 2 with MU-MIMO provides modern high-speed wireless connectivity
- IEEE 802.3af PoE powered eliminates need for AC power outlets at ceiling locations
- Dual-band operation ensures backward compatibility with legacy 2.4 GHz devices
- Fast roaming (802.11r/k/v) enables seamless handoff as users move between APs
- Enterprise-grade security (WPA2/WPA3) protects wireless network
- Low-profile design blends into office ceiling environment

4.c Uninterruptible Power Supplies: Specifications of selected products

Table - UPS Deployment and Specifications

Location	UPS Model	Capacity	Estimated Protected Load	Backup Target	Purpose
Server Room, RACK-GF (GF)	APC Smart-UPS SMT1500RM2U	1500 VA / 1000W	700W (ISR Router 250W + Core-Switch 95W + Switch-GF 95W + SW-Server Room 60W + Server 200W)	12-15 min - 70% load	Protects critical ground floor network infrastructure (router, core switch, distribution switch, server) during power outages, provides time for graceful shutdown or ride-through short outages
Conference Room, RACK-FF (FF)	APC Smart-UPS SMT1000RM2U	1000 VA / 700W	275W (Switch-FF 95W + SW-Managers 60W + SW-Conference 60W + SW-Dev 60W)	15 min - 40% load	Protects first floor network switches during power outages, maintains first floor connectivity if ground floor power fails temporarily

UPS Summary:

- **Total UPS Capacity:** 2500 VA / 1700W combined
- **Total Protected Load:** 975W
- **Combined Runtime:** 12-15 minutes average (sufficient for graceful shutdown or short outage ride-through)

APC Smart-UPS SMT1500RM2U

Technical Specifications

Model: SMT1500RM2U

Feature	Specification
Output Capacity	1500 VA / 1000W
Topology	Line-interactive with Automatic Voltage Regulation (AVR)
Waveform	Pure sine wave output (on battery mode)
Input Voltage	230V AC nominal, 160-286V AC operating range
Input Frequency	50/60 Hz auto-sensing ± 3 Hz
Output Voltage	230V AC ± 3% (on battery)

Feature	Specification
Transfer Time	4 ms typical (10 ms maximum, zero-downtime for IT equipment)
Output Receptacles	8× IEC C13 outlets (10A each)
Battery Type	Sealed Lead-Acid (SLA), maintenance-free, hot-swappable
Runtime with 50% Load (500W)	~20 minutes
Runtime with 70% Load (700W)	~12-15 minutes
Recharge Time	3 hours to 90% capacity
Battery Life	3-5 years typical (user-replaceable)
Surge Protection	Multi-stage surge protection, 480 Joules
Communication	USB, serial (RS-232), SmartSlot (optional network card)
LCD Display	Yes (real-time: load %, runtime remaining, battery %, input/output voltage)
Dimensions (H×W×D)	88mm × 432mm × 660mm (2U rackmount)
Weight	29 kg (with batteries)
MTBF	500,000 hours
Warranty	2 years standard

Justification for Selection:

- 1000W capacity adequate for ground floor critical load (700W)
- 12-15 min runtime allows graceful shutdown or ride-through common short outages in Sri Lanka
- Pure sine wave output essential for sensitive networking equipment (routers, switches)
- AVR automatically stabilizes voltage fluctuations without using battery (extends battery life)
- Hot-swappable battery enables zero-downtime battery replacement
- LCD display provides real-time monitoring without external software

APC Smart-UPS SMT1000RM2U

Technical Specifications

Model: SMT1000RM2U

Feature	Specification
Output Capacity	1000 VA / 700W
Topology	Line-interactive with AVR
Waveform	Pure sine wave

Feature	Specification
Runtime with 40% Load (275W)	~15 minutes
Output Receptacles	6× IEC C13 outlets
Battery Type	Sealed Lead-Acid, hot-swappable
Dimensions (H×W×D)	88mm × 432mm × 482mm (2U)

Justification for Selection:

- Right sized for first floor switches (275W = 39% utilization, excellent headroom)
- 15 min runtime sufficient for graceful shutdown if ground floor power fails
- Same enterprise features as SMT1500 (AVR, pure sine wave, LCD display)
- Compact 2U design fits in 12U wall-mounted RACK-FF

4.d Router: Specifications of selected product

Table - Router Deployment and Specifications

Item	Model	Ports	Link Type	Purpose
Edge Router	Cisco ISR4331-V/K9	4× Gigabit Ethernet (3 WAN + 1 LAN flexible configuration)	Copper 10/100/1000BASE-T (RJ45)	Internet gateway connecting SLT Fiber 100 Mbps to internal network, performs NAT/PAT address translation, zone-based firewall security, DHCP server for internal clients, and VPN termination capability

Cisco ISR4331

Technical Specifications
Model: ISR4331-V/K9

Feature	Specification
Form Factor	2U rackmount, 19-inch EIA standard
Interfaces	4× 10/100/1000BASE-T Gigabit Ethernet ports (RJ45)
Throughput	Up to 100 Mbps with security features enabled
NAT/PAT	Yes (Network Address Translation for internal network)
Firewall	Zone-based firewall (Cisco IOS built-in)
VPN Support	IPsec VPN, up to 50 Mbps VPN throughput
DHCP Server	Yes (automatic IP assignment for LAN clients)
Routing Protocols	Static routes, RIP, EIGRP, OSPF, BGP

Feature	Specification
QoS	Hardware-based Quality of Service, 8 queues per interface
IEEE 802.1Q	VLAN tagging support on all ports
Management	CLI (console/SSH), Web GUI, SNMP v3, Syslog
Memory	4 GB DRAM, 8 GB Flash storage
Expansion	3× NIM slots for future modules (4G backup, additional ports)
Power	250W maximum consumption, AC 100-240V 50/60 Hz
Dimensions (H×W×D)	88mm × 438mm × 406mm (2U)
MTBF	150,000 hours

Justification for Selection:

- 100 Mbps throughput perfectly matches SLT Fiber 100 Mbps internet service
- Built-in firewall and VPN eliminate need for separate security appliance
- NAT/PAT enables entire internal network (192.168.x.x) to share single public IP
- DHCP server automates IP address management for 40+ devices
- 3 NIM slots allow future expansion (4G backup WAN, additional interfaces)
- Enterprise-grade reliability with 150k hours MTBF

Router Configuration in Network:

- **WAN Interface (Gi0/0/0):** Connected to SLT ONT (100 Mbps fiber handoff)
- **LAN Interface (Gi0/0/1):** Connected to Core-Switch-01 (1 Gbps copper)
- **Services Enabled:** NAT, DHCP server (192.168.0.1 gateway), zone-based firewall, default route to ISP

Active Components Summary

Table - Complete Active Equipment Inventory

Category	Model	Quantity	Power per Unit	Total Power
Router	Cisco ISR4331	1	250W	250W
Layer 3 Switches	Catalyst 3650-24PS	3	95W (without PoE load)	285W
Layer 2 Switches	Catalyst 2960-24TT	6	60W	360W
Wireless APs	Cisco Aironet 1832i	5	12.5W (via PoE)	62.5W
UPS (Large)	APC SMT1500RM2U	1	700W load protected	-
UPS (Small)	APC SMT1000RM2U	1	275W load protected	-

Category	Model	Quantity	Power per Unit	Total Power
Total Active Devices		**16 devices**		**957.5W total load**

Key Compliance Points:

- **IEEE 802.1Q VLAN Support:** All 9 switches support VLAN tagging (assignment requirement met)
- **Wired Device Capacity:** 244 switch ports available for 41 wired devices (583% capacity, 75% expansion)
- **Wireless Device Capacity:** 5 APs support 60+ concurrent clients (exceeds 32 minimum requirement)
- **Power Protection:** 1700W UPS capacity protects 975W critical load (57% utilization, adequate headroom)

5 Cost Estimate

Pricing Sources

Supplier	Products	Website
Cisco Products Co (Pvt) Ltd	Cisco equipment (routers, switches)	www.optimatec.lk
Ubiquiti Products SL	Ubiquiti access points and switches	www.ubiquiti.lk
Singer Sri Lanka	APC UPS units	www.singer.lk
East Link Engineering Co (Pvt) Ltd	Cables, racks, patch panels	www.eastlink.lk
T&G Genex (Lanka)	Fiber optic cables and components	www.tggenex.lk
SLT (Sri Lanka Telecom)	Internet service	www.slt.lk

5.a Active Network Equipment Costs

Table - Active network equipment costs

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Router	Cisco RV340 Dual WAN Router	1	135,000	135,000	VPN, firewall, 4× Gigabit ports. Source: Cisco Products Co (Pvt) Ltd (www.optimatec.lk)
PoE Switch (Core/Distribution)	Ubiquiti UniFi Switch 24 PoE (250W)	3	98,000	294,000	24-port Gigabit PoE+, 250W budget, 802.1Q VLAN. Source: Ubiquiti

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
					Products SL (www.ubiquiti.lk)
Access Switch (Non-PoE)	Ubiquiti UniFi Switch 24 (Non-PoE)	6	45,000	270,000	24-port Gigabit, 802.1Q VLAN capable. Source: Ubiquiti Products SL (www.ubiquiti.lk)
SFP Fiber Module	Generic 1G SFP LC module	6	9,500	57,000	For fiber uplinks between floors (1 Gbps). Source: East Link Engineering (www.eastlink.lk)
Wireless AP	Ubiquiti UniFi 6 Lite	5	35,000	175,000	Wi-Fi 6 (802.11ax), PoE powered, dual band. Source: Ubiquiti Products SL (www.ubiquiti.lk)
AP Mounting Kit	Ceiling mount brackets	5	1,500	7,500	T-bar ceiling clips. Source: East Link Engineering (www.eastlink.lk)
UPS (Large)	APC Smart-UPS SRT 2000VA RM	1	195,000	195,000	2000VA/1600W online UPS, 2U rackmount. Source: Singer Sri Lanka (www.singer.lk)
UPS (Small)	APC Smart-UPS 1000VA RM	1	98,000	98,000	1000VA/700W online UPS, 2U rackmount. Source: Singer Sri Lanka (www.singer.lk)
Spare UPS Battery	APC Replacement Battery Cartridge	1	22,000	22,000	For 2000VA UPS unit. Source: Singer Sri Lanka (www.singer.lk)
Subtotal				**1,253,500**	

5.b Passive Network Components Costs

Table - Passive network components costs

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Equipment Racks					
Floor Rack	42U, 600×1000mm, 4-post	1	82,000	82,000	For RACK-GF ground floor. Source: East Link Engineering (www.eastlink.lk)
Wall Rack	12U, 400×700mm, wall-mount	1	32,000	32,000	For RACK-FF first floor. Source: East Link Engineering (www.eastlink.lk)
PDU (Vertical)	12-outlet rackmount	2	7,500	15,000	For RACK-GF (2 units). Source: East Link Engineering (www.eastlink.lk)
PDU (Horizontal)	8-outlet rackmount	1	5,500	5,500	For RACK-FF. Source: East Link Engineering (www.eastlink.lk)
Cable Management	1U horizontal (10 pcs)	10	1,100	11,000	Horizontal cable organizers. Source: East Link Engineering (www.eastlink.lk)
Mounting Hardware	Cage nuts, screws, washers	1 set	3,200	3,200	M6 cage nuts and rack hardware. Source: East Link Engineering (www.eastlink.lk)
Patch Panels					
UTP Patch Panel	24-port CAT6, 1U, T568B	4	7,800	31,200	PP-GF-01, PP-GF-02, PP-FF-01, PP-FF-02. Source: East Link Engineering (www.eastlink.lk)
Fiber Patch Panel	12-port LC duplex, 1U	2	11,000	22,000	FP-GF-01, FP-FF-01, OM3 multimode. Source: T&G Genex (www.tggenex.lk)

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Structured Cabling					
CAT6 UTP Cable	305m box, CMR-rated	3 boxes	26,000	78,000	915m total for horizontal cabling. Source: East Link Engineering (www.eastlink.lk)
OM3 Fiber Cable	6-strand, OFNR, aqua jacket	35 m	750	26,250	Inter-floor backbone fiber, pre-terminated. Source: T&G Genex (www.tggenex.lk)
RJ45 Connectors	CAT6 shielded (bag of 100)	1 bag	5,500	5,500	For field termination. Source: East Link Engineering (www.eastlink.lk)
Fiber LC Connectors	Pre-terminated pigtails (12pc)	1 set	14,000	14,000	For patch panel termination. Source: T&G Genex (www.tggenex.lk)
Cable Ties	Velcro, 100 pcs	2 packs	1,400	2,800	Cable management. Source: East Link Engineering (www.eastlink.lk)
Cable Labels	Vinyl, permanent (200 pcs)	1 set	4,200	4,200	Port and cable documentation. Source: East Link Engineering (www.eastlink.lk)
Wall Outlets					
Dual-Port Outlet	2× RJ45, faceplate + jacks	23	1,100	25,300	Workstation outlets. Source: East Link Engineering (www.eastlink.lk)
Single-Port Outlet	1× RJ45, faceplate + jack	8	750	6,000	Printer/dedicated outlets. Source: East Link Engineering (www.eastlink.lk)
Ceiling Outlet Box	Surface-mount box + jack	5	1,400	7,000	For wireless APs (PoE). Source: East Link Engineering (www.eastlink.lk)

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Floor Box (Conference)	2-gang floor box + 2 jacks	1	8,000	8,000	Conference table connectivity. Source: East Link Engineering (www.eastlink.lk)
Mounting Boxes	Single-gang plastic boxes	40	140	5,600	Wall outlet back boxes. Source: East Link Engineering (www.eastlink.lk)
Cable Tray & Conduit					
Cable Tray	100mm wire mesh, 3m sections	16 sections	2,000	32,000	Hallway cable management (48m total). Source: East Link Engineering (www.eastlink.lk)
Tray Hangers	Ceiling support brackets	32	380	12,160	Support every 1.5m. Source: East Link Engineering (www.eastlink.lk)
PVC Conduit (2")	Schedule 40, 3m lengths	4 lengths	1,700	6,800	For horizontal cable bundles. Source: East Link Engineering (www.eastlink.lk)
PVC Conduit (1")	For fiber, 3m lengths	9 lengths	1,100	9,900	Fiber riser conduit (27m total). Source: T&G Genex (www.tggenex.lk)
Conduit Fittings	Elbows, couplings, junction boxes	1 lot	7,800	7,800	Assorted PVC fittings. Source: East Link Engineering (www.eastlink.lk)
Fire-Stop Putty	For wall penetrations	2 kg	2,300	4,600	Fire-rated sealant. Source: East Link Engineering (www.eastlink.lk)
Subtotal				**457,810**	

5.c Internet Service Provider (ISP) Costs

Table - Internet service provider (ISP) costs

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Installation Fees					
Fiber Installation	GPON FTTH setup	1	15,000	15,000	One-time installation fee. Source: SLT (www.slt.lk)
ONT Deposit	Refundable equipment deposit	1	10,000	10,000	Refundable upon service termination (excluded from project total). Source: SLT (www.slt.lk)
Monthly Service (Year 1)					
Monthly Subscription	100 Mbps symmetric fiber	12 months	12,000	144,000	SLT Fiber Business 100 package. Source: SLT (www.slt.lk)
Static IP Fee	Public IPv4 address	12 months	1,500	18,000	Monthly add-on for static IP address. Source: SLT (www.slt.lk)
Subtotal (First Year)				**177,000**	
Next Years	Monthly fees only (12 months)	Per year	13,500	**162,000**	

Note: ONT deposit (LKR 10,000) is refundable and excluded from project total cost calculation.

5.d Cost of Installation Labor and Services

Table – Cost of Installation labor and services

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Structured Cabling	CAT6 installation + termination	37 outlets	2,400	88,800	Pull cables, terminate jacks, test connections. Local certified contractor
Fiber Installation	OM3 inter-floor + termination	1 run	24,000	24,000	Fiber pull through conduit, termination, testing. T&G Genex certified installer
Rack Installation	Floor rack assembly + leveling	1 unit	7,500	7,500	RACK-GF installation and leveling. Local contractor
Wall Rack Installation	Wall-mount + concrete anchoring	1 unit	11,000	11,000	RACK-FF wall mount with M12 anchors. Local contractor
Cable Tray Installation	Tray + hanger mounting (48m)	48 m	380	18,240	Ceiling-mount cable tray system. Local contractor
Equipment Installation	Rack equipment + patch cabling	1 lot	32,000	32,000	Install switches, routers, UPS, make all connections. Local contractor
Testing & Certification	CAT6 + fiber testing	1 lot	26,000	26,000	Cable certification with Fluke tester, test reports. Local contractor
AP Installation	Ceiling mount + basic config	5 units	3,800	19,000	AP mounting and basic wireless config. Local contractor
Documentation	As-built drawings + labeling	1 lot	14,000	14,000	Floor plans, cable schedule, outlet/patch labels. Local contractor
Subtotal				**240,540**	

5.e Miscellaneous and tools cost

Table - Miscellaneous and tools cost

Category	Item/Model	Qty	Unit Price (LKR)	Total (LKR)	Notes
Punch-Down Tool	110-blade punch tool	1	3,200	3,200	For patch panel termination. Source: East Link Engineering (www.eastlink.lk)
Cable Tester	CAT6 certifier (weekly rental)	1 week	7,500	7,500	Fluke cable tester rental. Local tool rental company
Fiber Cleaner	LC connector cleaning kit	1	4,200	4,200	Fiber optic cleaning supplies. Source: T&G Genex (www.tggenex.lk)
Drill & Bits	Masonry drill for wall penetrations	1 set	11,000	11,000	Professional grade drill and bits. Local hardware supplier
Ladder (Rental)	3m step ladder, 2 weeks	1	4,500	4,500	For ceiling work and AP installation. Local tool rental
Safety Equipment	Hard hats, gloves, safety glasses	1 set	5,500	5,500	PPE for installation crew. Local supplier
Subtotal				**35,900**	

5.f Project Cost Summary

Table - Project Cost Summary

#	Cost Category	Subtotal (LKR)	% of Subtotal
1	Active Network Equipment	1,253,500	57.9%
2	Passive Network Components	457,810	21.1%
3	Internet Service Provider (ISP)	177,000	8.2%
4	Installation Labor and Services	240,540	11.1%
5	Miscellaneous and Tools	35,900	1.7%
	Subtotal	**2,164,750**	**100.0%**

Conclusion

This network design provides a reliable and scalable infrastructure for Software Solutions (Pvt) Ltd. It effectively supports 41 wired and over 60 wireless devices across two floors. By using local Sri Lankan suppliers and modern equipment, the solution ensures high performance, security, and room for future growth, all while keeping costs down. VLAN-compliant switches, enterprise-grade wireless, and strong UPS protection ensure uninterrupted operations. The final cost is affordable, offering great value and a solid foundation for the company's IT needs in the coming years.